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Development of IoT based Smart Crop and Pest Management System for Sustainable Smart Farming

Sikha Nayak
Research Scholar
Dept. of CSE
Sangam University
Rajasthan
Bhilwara, India
shikha.nayak21@gmail.com

Awanit Kumar
Assistant Professor
Dept. of CSE
Sangam University
Rajasthan
Bhilwara, India
awanit.kumar@sangamuniversity.ac.in

Nirmal Singh
Assistant Professor
Department of CSE
Sangam University
Rajasthan
Bhilwara, India
nirmal.singh@sangamuniversity.ac.in

Ajay Kumar Suwalka
Assistant Professor
Dept. of CSE
Al Baha University
Alaqiq, Albaha
Saudi Arabia
ajay.suwalka@sangamuniversity.ac.in

Snehlata Ajmera
Assistant Professor
Dept. of CSE
Sangam University
Rajasthan
Bhilwara, India
sneha.maloo21@gmail.com

Shashank Sultaniya
Assistant Professor
Dept. of EE
Sangam University
Rajasthan
Bhilwara, India
Shashank.sultaniya@sangamuniversity.ac.in

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***Abstract*—"India suffers significant economic losses each year as a result of insects. The agriculture industry suffers enormous losses as a result of a secret swarm of insects that progressively harm plants. However, if farmers are unable to detect these insects in their early stages, they may use pesticides and insecticides without realizing their presence. This could therefore worsen the state of the economy. The design and development of an Internet of Things (IoT) based smart crop and pest management system is presented in this work. It includes an Arduino microcontroller, a GSM module, a fire sensor, an ultrasonic sensor, and a soil-moisture sensor. An ultrasonic sensor can detect insect sounds by listening for ultrasonic signals generated during insect feeding on crops. Once the existence of insects has been confirmed by an ultrasonic sensor, the farmer can next administer pesticides or insecticides according to the crop type. A GSM module will send a message alerting the farmer to the presence of insects. A fire sensor's primary function is to detect and respond to flames or fires. The level of soil moisture in the active root zone is measured by the soil moisture sensor (SMS), a sensor that is connected to the irrigation system controller. The GSM module will also be used to relay a message informing the farmer of the crops' soil-moisture condition. Besides helping farmers, this research increases agricultural productivity**

Keywords—: Smart farming; Internet of Things; Sensors; Early pest detection; Crop production; pesticides; Automatic irrigation.

I. Introduction

In addition to raising farmers' per capita income, vegetables also ensure the country's food and nutritional security, making them vital to Indian agriculture. With 16.7% of the world's total acreage and 15.4% of worldwide production, India ranks as the second-largest producer of vegetables. The primary global barriers to vegetable production are diseases and insect pests. The most common pests that cause substantial financial losses in the nation include blight, viruses, fruit flies, sucking pests, worms that develop knots in roots, mildews, and fruit flies. Climate change and agricultural progress have caused a paradigm shift in pest infestation in terms of time and location in recent years. An all-encompassing approach that integrates clever contemporary methods for vegetable protection is necessary to successfully handle these biotic problems. Lack of understanding often leads farmers to use chemical pesticides indiscriminately to control pests. The average crop losses in the country due to various pests range from 10 to 30 percent as mentioned in table 1. The problem is made worse by the fact that many of them not only directly injure people but also act as carriers of several viral illnesses. A variety of existing equipment can be enhanced with IoT and smart sensors to further increase agricultural output by facilitating the diagnosis of pest infestations and crop illnesses.

TABLE I. Yield losses due to major insect pests and diseases in major vegetables in India

Crop/Pest	*Yield loss (%)*	*Crop/Pest*	*Yield loss (%)*
Tomato		**Brinjal**	
Fruit borer (Helicoverpaarmigera)	25-65	Fruit and shoot borer (Leucinodesorbonalis)	11-94
Early blight (Alternaria solani)	Up to 79	Blight (Phomopsis vexans)	35-50
Wilt (Fusarium oxysporum f. sp. Lycopersici)	11 to 61	Root knot nematode (M. incognita, M. javanica)	16.72
Wilt (Verticillium dehalie)	25-30	**Cabbage**	
Begomoviruses	100	Diamond back moth (Plutellaxylostella)	18-99
Root knot nematode (Meloidogyne spp)	27.5	Caterpillar (Pieris brassicae)	67
Chilli		Leaf webber (Crocidolomiabinotalis	29-50
Thrips (Scirtothrips dorsalis)	12-95	Cabbage borer (Hellulaundalis)	35-55
Mites (Polyphagotarsonemus latus)	36	Black rot (Xanthomonas campestris pv. campestris)	51

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Crop/Pest	*Yield loss (%)*	*Crop/Pest*	*Yield loss (%)*
Anthracnose (Colletotrichum spp.)	32-82	Cucurbits	
Viral disease due CMV and PVY	66.8-75	Fruit fly (Bactrocera cucurbitae) Bitter gourd	62-82
Root knot nematode (M. incognita, M. javanica)	12.87	**Cucumber**	22-38
Okra		Ivy gourd	63
Fruit borer (H. armigera)	24	Musk melon	77-100
Leafhopper (Amrasca biguttulabiguttula)	55-67	Snake gourd	64
Whitefly (Bemisiatabaci)	55	Sponge gourd	51
Shoot and fruit borer (Eariasvittella)	24-54		
Bud rot (Choanephora cucurbitarium)	6-50		

Farmers may be able to better manage veggies and crops over a wider area in less time with the help of the Internet of Things and a variety of smart sensors. Additionally, the Internet of Things might facilitate the process of deciding when and why to use pesticides and fertilizer in the field. Excessive use of pesticides and fertilizers can harm the environment and endanger human health. Without a doubt, the Internet of Things may reduce resource waste and safeguard farmers' income.

IoT and Arduino sensors can be used to collect precise data on temperature, moisture content, soil quality, and insect infestation in a particular area. Smart sensors based on the Internet of Things can identify a variety of pests in agricultural fields through the emission of ultrasonic sound waves. Most animals can generate and hear ultrasonic frequencies, but humans cannot sense them. Many animals, such as insects, bats, and rodents, also communicate using these frequencies. In insects like flies and spiders, the "tympanic membrane" detects ultrasonic frequencies. Cockroaches have "sensory hairs," which enable them to sense. The following outlines how animals react to ultrasonic frequencies: -

- 22–25 kHz: Cats and Dogs
- House flies, spiders, and fleas at 38–44 kHz
- 52–60 kHz: Reptiles
- Rats at 60–72 kHz

Even though ultrasound has been shown to be biologically safe for people, children are able to perceive frequencies that are closer than 30 kHz. To detect insect activity, developed system can listen for insect sounds in the 20–20,000 Hz audio frequency range. Considering chemical pesticides have many detrimental impacts on beneficial creatures, these technologies are a good alternative. Crop monitoring has changed from being a qualitative, experience-based endeavour to a quantitative, data-driven one because of the adoption of IoT for soil, microclimate, and crop sensing. IoT-based technology allows farmers to precisely monitor the development and health of their crops. Agricultural personnel can also use it to quickly identify pest assaults and plant diseases.

Sensors placed in the field to gather information on a range of environmental characteristics, such as temperature, humidity, soil moisture, and nutrient levels, are essential parts of Internet of Things-based smart crop and pest management systems. Due to their frequent networking, these sensors may transmit data in real time to a centralized cloud platform. To understand more about crop health, growth rates, and yield potential, the data collected by these sensors is subsequently subjected to machine learning and other data analytics techniques. Farmers can use this information to make decisions regarding crop management, including harvesting, pest control, and irrigation.

II. Background Study

Smart farming using IoT in a systematic manner using the recommended reporting items for systematic reviews (PRISMA) methodology [1]. The review's objectives are to list the principal tools, platforms, network protocols, data processing technologies, and ways that smart farming and the Internet of Things can benefit the agricultural sector.

The current study attempts to elucidate possible effects of pesticides on various organism taxa and agricultural lands. PRISMA guidelines were used for this purpose [2]. Literature from the Science Direct and PubMed databases was screened using a variety of search phrases. As a result, the current work offers a thorough analysis of the harmful effects of pesticides on biotic biosphere components.

The goal of agricultural precision systems is to handle agricultural monitoring and actuation in a better, simpler, more affordable, and generally more efficient manner [3]. More precisely, author suggested combining all of these technologies in this study to create a precision agriculture system prototype for medium-sized and small plantations that emphasizes only two major key benefits: economic policy and effective energy management with self-charging capabilities [4].

Calibration and identification of sensor failures is a critical problem for Internet of Things applications [5]. We suggest the Calibration Talk method as a solution to this problem. We demonstrate how Calibration Talk automatically resolves sensor accuracy issues in commercial agricultural operations using smart agriculture as an example. In addition to variations in manufacturing, hardware failure, and age, soil erosion from irrigation or rain can also lead to sensing inaccuracy by altering the locations and angles of sensors placed in the soil.

The goal of this project is to maximize water utilization for irrigation farming by designing an affordable, autonomous sensor interface that can automate the monitoring and control of irrigation systems in remote places [6]. The goal of a smart irrigation system is to enable the autonomous delivery of sufficient water from a reservoir to domestic crops through the use of sensors and actuators in an internet of things-based irrigation monitoring and control system.

In this paper, we simply use several sensors to recognize

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insect behavior to provide an approach for pest monitoring using wireless sensor networks [7]. Based on five significant crops and the related insect pests, we suggested a quick and precise method for identifying and classifying insects. This approach gathers information from field-installed sensors to study insect behavior.

The aim of this research is to develop a real-time monitoring system for insect pest traps based on the Internet of Things (IoT) by utilizing sensors [8]. Additionally, the study aims to determine the advantages of IoT technology in this regard. The ESP32-WROOM microcontroller, pest sting module, ultrasonic sensor, DHT11 sensor, passive infrared receiver sensor, and Blynk platform are all used in the automatic trap and midges monitoring system [9]. core of our implemented system is the smart crop and pest management solution, which is essential for implementing various effective IoT-based strategies and methods to prevent pest threats over time. IoT-based smart crop and pest management systems use a range of tools from the mechanical, biological, chemical, and cultural domains. It will help farmers and agricultural researchers to manage pests and boost fabricating gain [10].

TABLE II. COPARATIVE ANALYSIS WITH EXSHISTING METHOD

Feature / Study	IoT-Based Irrigation	Pest Monitoring	Fire Detection	Ultrasonic Sensing	GSM Alert System	Reference
Bhalekar et al. (2023)	Yes	No	No	No	No	[11]
Rajak et al. (2023)	No	Yes (with ANN)	No	No	No	[12]
Patel et al. (2024)	No	Yes	No	Yes (Pest trap)	No	[13]
Morchid et al. (2024)	No	No	Yes	No	Yes	[14]
Bhalekaret et al. (2023)	Yes	Yes (Sprayer-based)	No	Yes	No	[15]
Proposed Model	Yes	Yes	Yes	Yes	Yes	This work

III. PROPOSED METHODOLOGY

The core of our implemented system is the smart crop and pest management solution, which is essential for implementing various in Fig.1 effective IoT-based strategies and methods to prevent pest threats over time. IoT-based smart crop and pest management systems use a range of tools from the mechanical, biological, chemical, and cultural domains. It will help farmers and agricultural researchers to manage pests and boost fabricating gain.

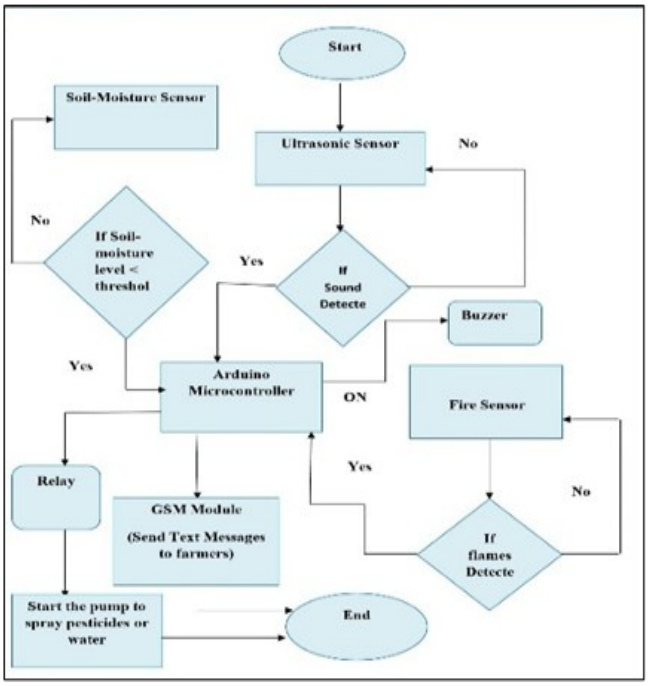

Fig. 1: Flow Chart of Implemented System Methodology

IV. IMPLEMENTATION

Design Units covers all the items like microcontroller, sensors, modules used to develop this research project. Fig.1 depicts the architecture diagram of the system we developed and put into use.

Ultrasonic sensors convert ultrasonic waves into electrical impulses. Transducers that can send and receive ultrasonic waves are known as ultrasound transceivers. Ultrasonic waves, which humans can hear over specific thresholds, are produced by sound waves. To detect insect activity, the implemented system listens for insect sounds in the 20–20,000 Hz audio frequency range. To identify insects, the "acoustic emission" technique is useful. Non-destructive testing of metal assemblies and parts has been the primary use of this technique.

Farmers can identify eating insects that emit ultrasonic signals by employing acoustic emission technologies. When an insect tears the fibrous tissue it consumes, mechanical disruptions occur throughout the food source. The acoustic emission technique detects these ultrasonic waves that are produced by these mechanical disturbances. A "feeding event" will be characterized as taking a "bite" out of the content. The majority of human activities produce very little acoustic energy at ultrasonic frequencies. As a result, the noises that the transducer picked up were those produced by the tearing off of plant tissue during insect eating. The ultrasonic mechanical disturbances are directly linked to the insect's mouthparts ripping the material.

An essential part of the implemented systems for identifying flames is a flame/fire sensor. The ability to detect the unique characteristics of flames, such as light, infrared radiation, or ultraviolet radiation, depends on the type of flame sensor. It can be positioned strategically throughout the farm field, and It will notify a message alert and sound alarm upon detecting smoke or fires. This makes it possible for crop

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field fires to be detected early and responded to swiftly, enabling farmers to act swiftly and reduce the amount of damage caused by wildfires.

To detect the quantity of moisture in the soil in the active root zone, the irrigation system controller is connected to a sensor called the soil moisture sensor (SMS). The sensor comprises two exposed conductors and a fork-shaped probe that may pierce soil Showsin Fig.2 The sensor is coupled to an electrical module that transforms probe signals into analog and digital values that can be read by an Arduino microcontroller. If the soil moisture level drops below a preset threshold, an Arduino will trigger a relay to turn on the pump and irrigate the plant. When the soil moisture level rises above a preset threshold, an Arduino will trigger a relay to stop the pump and watering illustrates in Fig.3.

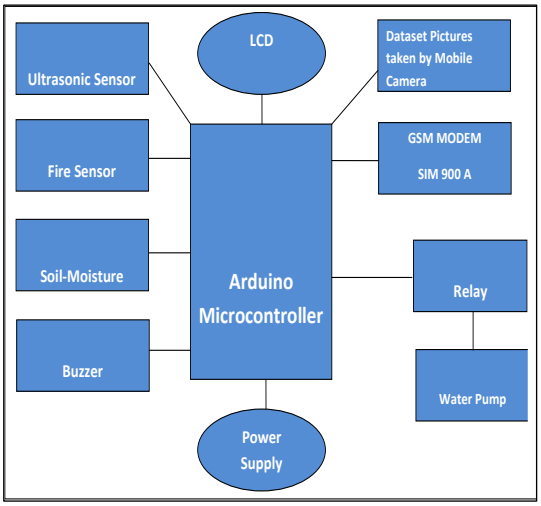

Fig.2 : Architecture Diagram of Implemented System

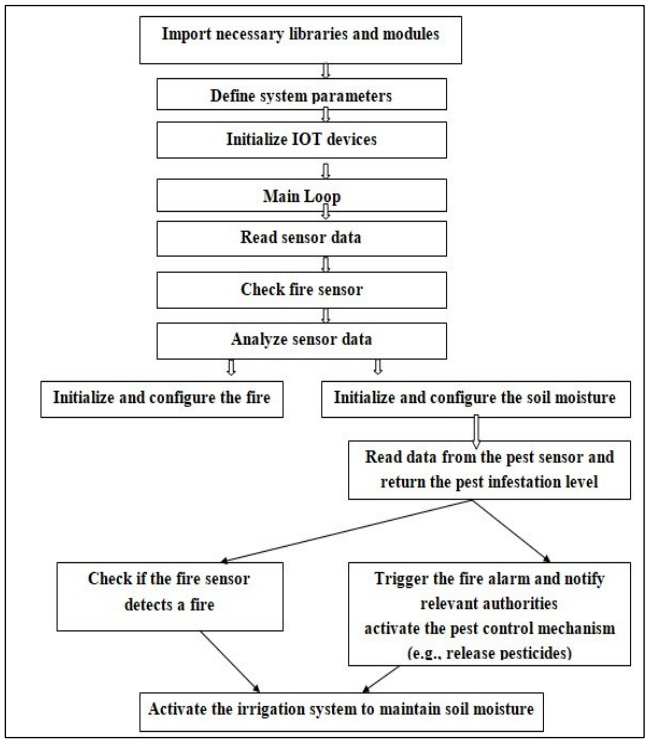

Fig.3: Working of IoT Based Crop And Pest Management System

V. Result And Discussion

To accomplish the goals of this research study and obtain more precise and accurate values, few tests have been carried out in this part. The final implemented system is illustrated in fig.3. Testing of IoT based crop and pest management system has been shown in fig.4. As seen in the figure below, we put the entire circuit—aside from the pump and sensor probe-into a plastic box after uploading the code to the Arduino. Insert the moisture sensor into the soil in Fig.4&5 For best accuracy, place the sensor as close to the plant roots as you can.

To test the ultrasonic sensor's ability to disrupt and repel pests that perch on vegetable crops, ultrasonic waves were emitted. A single ultrasonic sensor used in this study. To determine whether the sensor is successfully keeping insect pests away from vegetable plants, this test was repeated multiple times by measuring the output voltage on the sensor's trig pin. By generating ultrasonic waves as midges, ultrasonic sensors can function effectively, according to the data in Table 2. However, "passive" pests which have a propensity to remain silent and perch on vegetable plants moved differently when the radius of the pests was tested from 0.5 meters to 2.5 meters. A propensity to depart the vegetable plant was noted in the movement of "active" pests during experiments conducted within a 3-to-4-meter radius.

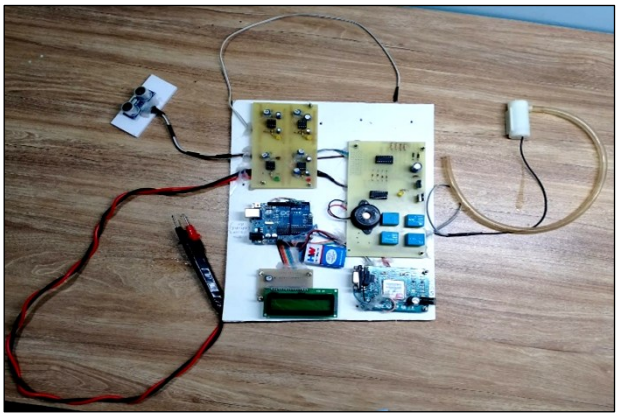

Fig.4 : Final implemented device

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TABLE III. VARIATION OF CLASSIFICATION ACCURACY

No.	*Distance (m)*	*Voltage (VDC)*	*Description (Active/Passive)*
		Ultrasonic	
1.	0,5	4,91	Passive
2.	1	4,87	Passive
3.	1,5	4,86	Passive
4.	2	4,85	Passive
5.	2,5	4,82	Passive
6.	3	4,83	Active
7.	3,5	4,80	Active
8.	4	4,87	Active

A flame/fire sensor is designed to identify light within a particular wavelength range. The sensor tracks the visible spectrum. According to specifications, a detection range of up to 33 cm is permissible. Changes occur in wavelength and distance values; the increase in nm is proportional to the distance. The accuracy increases with the distance. Reliability increases with decreasing distance. The reading in 5 centimetres is 26, far from 100; fewer than 100 fire is found. In 30cm, the reading is 97, which is critically close to 100, and sensors fluctuate within close range, so it can go more over 100 in thunder instant and not detecting fire due to sensitivity. Fig. 4 illustrates the fire sensor detection result. Table 3 shows that six different testing methods were used in an ongoing effort to track changes in the sensor. There have been six more readings to determine firefighting. The fire/flame sensor was initially in contact with flame. However, because of the reaction time, no fire was found. Every five seconds, the testing is conducted continuously and analyzed. A water sprinkler that stops when the fire sensor reaches 1024 was used to combat the fire in tests 2, 3, and 4.

TABLE IV. FIRE SENSOR TEST RESULT DATA

No.	*Fire Sensor Reading*	*Status*	*Buzzer*	*Water Sprinkler*	*Response Time*
1.	1024	No Fire	OFF	OFF	None
2.	34	Fire	ON	ON	2 Seconds
3.	22	Fire	ON	ON	3 Seconds
4.	10	Fire	ON	ON	2 Seconds
5.	1024	No Fire	OFF	OFF	4 Seconds
6.	1024	No Fire	OFF	OFF	None

Below Table 4 shows the threshold value for soil-moisture sensor. In this testing, different soil type, such as soil and sand, were used to determine the effect of sensor reading. Figure 5 illustrates the comparison between amount of moisture soil in the sand and soil. Table 5 illustrates the test result data of soil-moisture sensor for sand and soil

TABLE V. THE THRESHOLD VALUE AND STATUS OF SOIL-MOISTURE SENSOR

Threshold Values	*Status*
0%	Dry
> 0-25%	Rather Moist
26-50%	Moist
51-75%	Very Moist
76-99%	Wet

TABLE VI. TEST RESUL DATA OF SOIL-MOISTURE SENSOR FOR SOIL AND SAND

Time	*Sensor Reading (Soil)*	*Sensor Reading (Sand)*	*Condition*
14:10	0%	1%	Dry
14:20	32%	16%	Dry
14:30	51%	21%	Moist
14:40	77%	34%	Wet
14:50	95%	50%	Very Wet

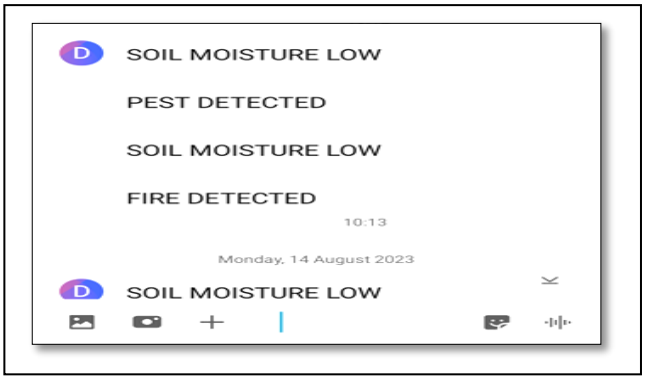

Fig.5 : Software Result of Iot based crop and pest management system

VI. CONCLUSION & FUTURE SCOPE

IoT-powered smart farming offers promising future opportunities. Modern sensors can monitor temperature, moisture content, and rainfall to optimize agricultural productivity. Every mechanical system has the capacity to protect us from the unnecessary use of pesticides and unpredictable agricultural irrigation, as well as to instantly alert farmers when cutting-edge applications are required to increase crop output. An ultrasonic sensor helps detect pest infestations early so the farmer can apply pesticides and insecticides as needed by sending a warning message to his registered mobile number. Therefore, this research helps alert farmers to increase the production of vegetable crops. Despite these advantages, a major obstacle to the broader adoption of IoT-based smart crop and pest management systems in agriculture is the cost of acquiring and maintaining advanced hardware and software. Additionally, rural farmers lack sufficient knowledge about how to operate IoT devices. Finally, cybercriminals may have a larger effect on IoT-based smart farming by damaging cloud servers that hold important data. Government programs encouraging digital literacy, data encryption, and financial support for farmers could help smart farming become widely adopted.

The primary goal of this research project is to inform farmers on pests, pesticide use, soil moisture status, and other environmental parameters. An Arduino microcontroller is a central component of the system. According to recent technological advancements, there are a number of different implementations that can improve the efficiency and usability of the system: Soil water and nitrogen concentrations can be measured with IoT-enabled sensors. Furthermore, evapotranspiration rates can be efficiently monitored to improve crop health surveillance based on the CO2 level in agricultural lands. In bigger fields, a UAV equipped with a high-resolution camera and additional attachments can be used to take real-time pictures of plants and insects. In the future, our developed system can

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incorporate an artificial pore based on microfluidics to detect the presence of microorganisms in agrifoods. Interpreting data collected by various sensors can be done with deep learning techniques. We can add automated intelligent IoT functionalities to measure several important indicators, such as the Vegetation Health Index (VHI), Standardized Anomaly Index (SAI), and Evaporative Stress Index (ESI).By collecting crucial data, we may use satellites and GPS systems to precisely monitor field cultivation, water levels, and fertilizer needs in real time.

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